Run report: allocator confound study sessions, 2026-10-23 06:10 UTC

Descriptive run report for the 14 ecbench sessions of `research/ecbench_allocator_confound_20261023` (preregistered; interpretation is in `../../REPORT.md`, not here). Observations only; wall time is descriptive.

What ran

- **Commands.** `run_all.sh` (ABBA of `base` and `fix` per spec) and the three ablation runs in `logs/`; every session is `ecbench run --wait --allow-busy --cpus auto --quiet --spec SPEC --out sessions/NAME`. Specs: `docs/ecbench/specs/smoke-prime.json` (sha256 684c9aeb…79cc2, spec id ECS1h94f54998e76f) and `specs/claim-koblitz-r10.json` (sha256 c23b5202…52392, spec id ECS1hd5d9c5a020f7).
- **Commits and binaries.** merge-base `3a1ea541d964d6ef9099300a9ab512d55eaa073b` (binary a59c4014ff52, grading v2); branch `ecbench/fix-runner` source `63fede816dcacc012580ce37fd40b4ad850ad867` (binary 26f25d9fa47a, grading v3); three scratch ablations of the branch source (`BINARIES.txt`).
- **Host.** Shared 4-vCPU Xeon 2.10 GHz VM, class ECBENV3had995aa2df0f; the cargo lock was held by other agents at every session start, load average 2.7 to 5.1. `--allow-busy`, whole core reserved by `--cpus auto`; levels earned were L0 (two ablations), L1 and L2, never L3 (`REPORT.md`, "Design").
- **Arms, workloads, seeds.** smoke-prime: 9 arms (`rho-neg` reference, `rho-plain`, `rho-frozen`, `bsgs`, `bsgs-il`, `bsgs-neg`, `kangaroo`, `ic-mitm`, `rho-neg-aa` A/A control), one registered 20-bit prime-order curve, 2 targets (target seed 20261002), 3 measured rounds + 1 warm-up, session seed 1. claim-koblitz-r10: `rho-strong` (reference), `rho`, `ic` (`koblitz-orbit:divisor=0;1`, `mitm-frobenius:m=2`) on Koblitz a=1 n=17 (`icv1-f2m17-tm101-00378d4e`), 2 public targets (target seed 11), 10 measured rounds + 1 warm-up, session seed 5.

Runs requested and completed

session	session id	start UTC	duration	status	planned	records written	binary	grading	spec id
base-smoke	ECBS1hd3725d8a32c3	06:10:03	3 s	complete	72	72	a59c4014ff52	2	ECS1h94f54998e76f
fix-smoke	ECBS1hff17ab9636f0	06:10:16	5 s	complete	72	72	26f25d9fa47a	3	ECS1h94f54998e76f
fix-smoke-rev	ECBS1h5d29a35e9eba	06:10:23	5 s	complete	72	72	26f25d9fa47a	3	ECS1h94f54998e76f
base-smoke-rev	ECBS1h420293eaa9c5	06:10:30	3 s	complete	72	72	a59c4014ff52	2	ECS1h94f54998e76f
base-claim	ECBS1h7c0082ac7265	06:10:36	4 s	complete	66	66	a59c4014ff52	2	ECS1hd5d9c5a020f7
fix-claim	ECBS1h157a7553e392	06:10:42	5 s	complete	66	66	26f25d9fa47a	3	ECS1hd5d9c5a020f7
fix-claim-rev	ECBS1ha418537656ed	06:10:49	5 s	complete	66	66	26f25d9fa47a	3	ECS1hd5d9c5a020f7
base-claim-rev	ECBS1h4d78060e6148	06:10:56	3 s	complete	66	66	a59c4014ff52	2	ECS1hd5d9c5a020f7
fix-notrim-smoke	ECBS1h628195e31e14	06:39:19	5 s	complete	72	72	6fbdb7c8f352	3	ECS1h94f54998e76f
fix-notrim-claim	ECBS1hc3e67b6813c4	06:39:26	4 s	complete	66	66	6fbdb7c8f352	3	ECS1hd5d9c5a020f7
fix-notrim2-smoke	ECBS1h0aa59508bb08	07:15:51	5 s	complete	72	72	4e9927811f4d	3	ECS1h94f54998e76f
fix-notrim2-claim	ECBS1h9d777ef2e237	07:15:58	5 s	complete	66	66	4e9927811f4d	3	ECS1hd5d9c5a020f7
fix-noprobe-smoke	ECBS1h322f7c2815eb	07:42:48	4 s	complete	72	72	591b9a402a4d	3	ECS1h94f54998e76f
fix-noprobe-claim	ECBS1h11617fea463f	07:42:55	4 s	complete	66	66	591b9a402a4d	3	ECS1hd5d9c5a020f7

All 14 sessions completed with every planned execution recorded; every record `verified` (no failure, timeout or interruption to report). Raw records: `sessions/NAME/records.jsonl`; audit receipts: `audits/NAME.audit.json` (all `ecbench verify --replay-all --exit-code`: 0 problems).

Measured values

Medians of `time.solve_wall_ns` (smoke, ms) of verified measured records; `base` and `fix` pooled over their two sessions (n = 12 per arm), each ablation one session (n = 6). Spread: quartiles and extremes in `analysis/summary.json`; per-record dots in the graph.

arm	base	fix (trim + probe)	fix-notrim (probe only)	fix-noprobe (trim only)	fix-notrim2 (neither)
rho-neg	5.833	0.25	0.236	0.277	6.57
rho-plain	6.144	0.277	0.25	0.275	6.738
rho-frozen	6.091	0.33	0.281	0.271	6.594
rho-neg-aa	6.767	0.25	0.221	0.278	6.419
bsgs	5.392	0.186	0.188	0.212	6.461
bsgs-il	5.262	0.162	0.14	0.182	5.769
bsgs-neg	5.303	0.137	0.123	0.14	6.177
kangaroo	6.24	0.196	0.172	0.215	8.342
ic-mitm	1.891	1.83	1.866	1.88	1.906

claim-koblitz-r10 (us; `base` and `fix` n = 40 per arm, ablations n = 20):

arm	window	base	fix	fix-notrim	fix-noprobe	fix-notrim2
ic	solve	1412.7	1354.2	1342.1	1456.2	1518.4
ic	online	127.1	119.1	113.5	125.2	120.3
rho	solve	5886.9	249.1	198	296.4	5638.7
rho	online	5885.3	245.6	194.1	292	5635
rho-strong	solve	5658.4	148.7	147.3	170	5631.3
rho-strong	online	34.7	38	37.7	42.6	39.2

References and controls: `rho-neg` is the smoke reference, `rho-strong` the claim reference; the A/A pair `rho-neg` and `rho-neg-aa` medians differ by a factor 1.160 (base) and 1.003 (fix). Operation counts: identical across binaries for every record (`REPORT.md`, "Counts"); the section 5a band at 0, 0.1, 1 GAE puts `rho-strong` ahead of `ic` on claim-koblitz at every weight (S 4.004, 4.140, 5.361 against 16.852, 17.283, 21.160).

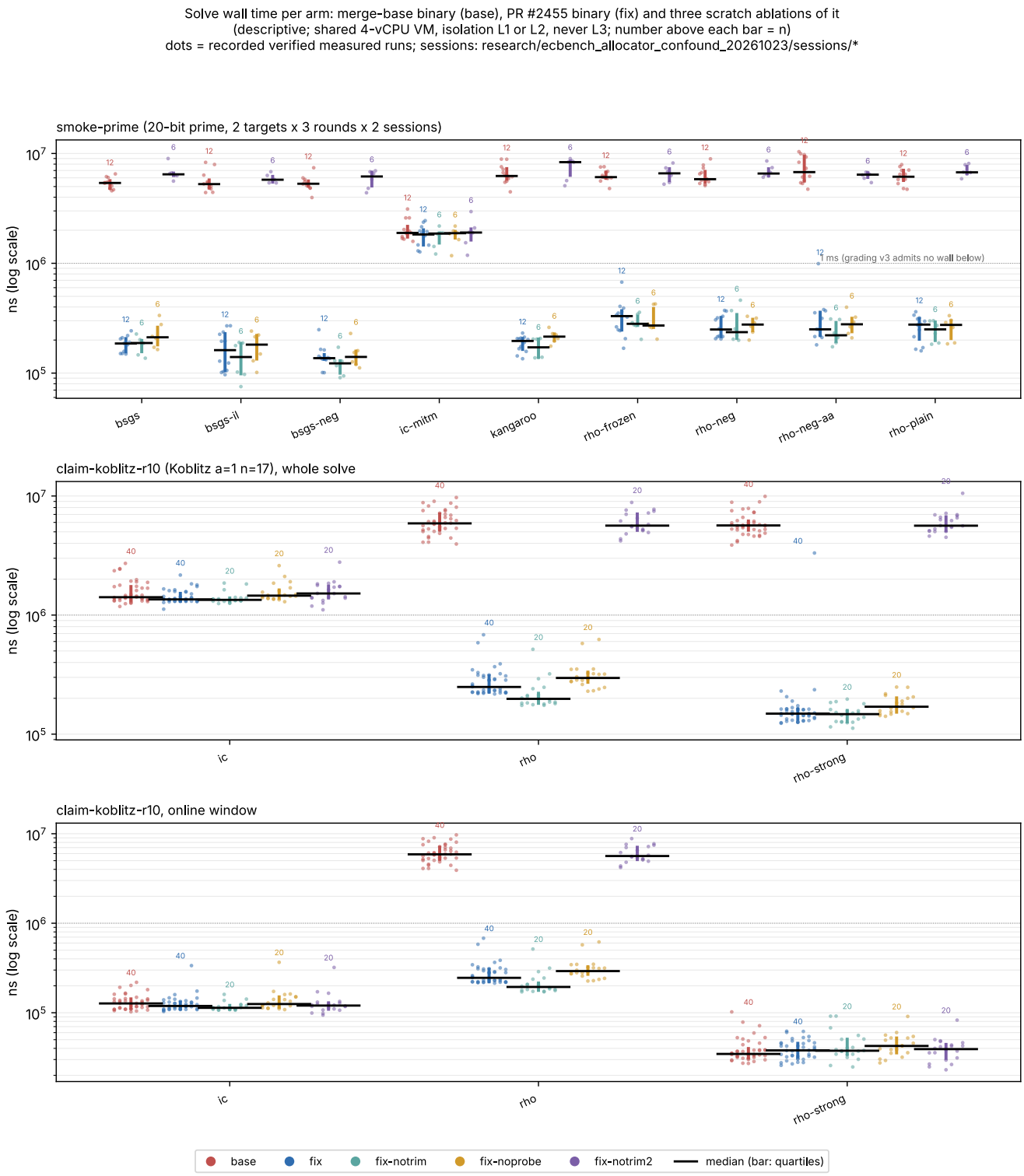


Figure 1: Recorded solve wall time per arm (dots, log scale), pooled median and quartile bar, n above each bar. Sessions `sessions/*`, run ids as in the table above; plotting source `analysis/plot_wall.py`, data `summary.json` beside this file.

Missing or not measured

No independent replay (same host, same agent), no L3 isolation, no hardware counters (`perf_event_open` unavailable on this VM), no trace of `malloc_consolidate` itself.